RoadGuard-AI: A Hybrid Edge–Cloud Cascaded Multi-Modal Framework for Real-Time Road Hazard Detection Using Sensor–Vision Fusion

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Abstract— Road surface irregularities, such as potholes and speed breakers, are identified as key factors in causing road accidents, road damage, and economic losses in urban and semi-urban areas. The existing methods in road inspection are found to be periodic, tedious, and inefficient in detecting dynamic or sparse road hazards. The existing AI-based approaches in road hazard detection are found to be unimodal, using only inertial sensors or computer vision data, and are found to be inefficient in generalizing diverse road scenarios in heterogeneous environments. This paper proposes a novel hybrid edge-cloud cascaded multi-modal road hazard detection framework, termed RoadGuard-AI, using a weighted late fusion of analysis results from tri-axial accelerometer data analysis and analysis results from computer vision data analysis using YOLOv8. A novel hierarchical two-stage cascaded framework is proposed in this paper, in which the three-class problem in road condition classification is transformed into a cascaded problem in presence/absence detection and type classification of road hazards. An exhaustive ablation study is performed to assess the impact of class balancing and data augmentation in cascaded and flat classification paradigms. In fact, the cross-validation results indicate a directional improvement in mean accuracy from 0.458 (flat) to 0.551 (cascaded), or directional improvement of 20.3% relative (p=0.107), warranting further evaluation with a larger dataset, albeit statistically insignificant (p = 0.107). For the baseline setting, where no data augmentation and class weighting are performed, the cascaded framework attains a peak accuracy of 0.725, whereas the flat model attains 0.673 in accuracy.Full-stack deployment architecture of the system includes a React Native mobile sensor acquisition layer, a FastAPI cloud inference backend, and an interactive Leaflet.js geospatial dashboard. RoadGuard-AI presents a research prototype toward a scalable and deployable civic intelligence system for proactive monitoring of road safety.

Keywords—*road hazard detection; sensor-vision fusion; cascaded classification; YOLOv8; accelerometer time-series; edge-cloud architecture; intelligent transportation systems.*

I. INTRODUCTION

Road infrastructure quality is a vital factor in ensuring traffic safety, the longevity of vehicles, as well as logistical efficiency. According to the World Health Organization, an estimate of 1.35 million fatalities occurs due to road traffic accidents every year, a considerable percentage of which are contributed to by poor road surface conditions, including potholed surfaces, undesignated speed breakers, as well as surface discontinuities [1]. However, despite the importance of this condition, the process of monitoring the condition of the majority of urban and semi-urban routes still remains a manually surveyed process, which is both temporally sparse, geographically incomplete, as well as operationally costly.

With the advent of embedded sensing systems and deep learning inference systems, there is a potential to leverage automated continuous detection of anomalies on the road. However, the majority of the systems are based on a unimodal approach, which includes the utilization of accelerometer threshold detection-based bump detection methods [2] or CNN/YOLO-based object detection methods for visual pothole detection [3]. These methods are restricted because of unimodal failure modes, where sensor-based systems are prone to false positives due to mechanical vibrations, while vision-based systems are prone to poor illumination conditions.

In this paper, we propose a novel hybrid edge-cloud multi-modal framework called RoadGuard-AI, which overcomes the limitations mentioned above through three main innovations: (1) the use of a hierarchical cascaded binary classification strategy; (2) the use of a probabilistic

late-fusion approach based on accelerometer model scores and YOLOv8 visual confidence scores; and (3) the use of a real-time full-stack architecture for deployment.

Main Contributions of This Work:

•A hybrid edge cloud multi-modal civic AI framework fusing tri-axial accelerometer time series data with YOLOv8 visual inference using a weighted probabilistic late fusion strategy for real-time road hazard classification.

•A hierarchical two-stage cascaded binary classification framework for decomposing the three-class problem of road anomaly detection into a hazard presence detection stage followed by a hazard type discrimination stage for achieving higher cross-validation accuracy compared to flat classification frameworks.

•A systematic ablation study on the interaction effects of class weighting and data augmentation strategies on cascaded as well as flat classification frameworks, along with the identification and resolution of a label encoding flaw compromising cross-validation integrity.

•A fully deployable end-to-end solution including a React Native mobile sensor acquisition module, a FastAPI cloud inference backend module, and a geospatial visualization module using Leaflet.js with spatial temporal deduplication support.

II. RELATED WORK

A. *Vision Based Road Anomaly Detection*

Initial studies on computer vision for automatic road surface inspection utilized handcrafted features like HOG and LBP with SVM classifiers for pothole boundary detection [4]. The use of deep CNNs significantly enhanced detection accuracy, and one-shot detection frameworks from YOLO achieved real-time detection of pothole locations with mAP > 0.60 on public benchmark datasets [5]. Accuracy is increased in YOLOv8 through the utilization of C2f modules, anchor-free heads, and detection paths. However, vision-based approaches have problems with regard to camera position, illumination, and occlusion.

B. Inertial Sensor-Based Detection

In accelerometer-based systems, vertical and lateral vibrations resulting from discontinuity in the road are utilized. A threshold-based approach sets a hazard flag when vertical acceleration is higher than a certain multiple of the running mean [6]. A statistical approach using tri-axial IMU data resulted in F1 scores in the range 0.78-0.85 [7]. Further improvement in sensitivity can be obtained by using LSTM networks for inertial sensor data. However, these systems lack spatial information required for differentiating structural from transient disturbances.

C. Multi-Modal Fusion Architectures

Sensor vision fusion in the context of road condition monitoring has been explored in prior studies. Late fusion approaches, in which each modality is independently scored before fusion, are found to be more robust in the presence of modality-specific sensor failures [7]. Dynamic fusion using attention-based cross-modal transformers, in which the temporal sensor modality is dynamically aligned with the visual modality, is found to achieve state-of-the-art performance on the benchmark [8], albeit at a computational cost that is currently infeasible for deployment in mobile contexts. RoadGuard-AI employs a probabilistic late fusion strategy, which is computationally efficient for mobile deployment.

D. Cascaded Classification Strategies

Cascaded classification, in which a coarse binary classification is performed before a detailed multiclass classification, has been found to be beneficial in domains in which class imbalance and interclass overlap are dominant characteristics [9]. In the context of road hazards, the use of cascaded classification is motivated by the fact that normal road segments and actual hazards are found to have similar low-frequency accelerometer signatures at low vehicle speeds, making three-class discrimination between them more error-prone**.**

E. Research Gap

The literature survey shows that existing systems face the following issues: (1) reliance on uni-modal inputs, (2) fixed classification architecture that does not account for hierarchical discrimination of hazards, (3) lack of ablation studies on regularization, and (4) lack of validation for deployment scenarios. The proposed RoadGuard-AI addresses all these issues in a unified framework.

III. PROPOSED METHODOLOGY

A. Problem Formulation

In the context of the provided set of data points $X = \{x_1, x_2, \ldots, x_n\}$, where the individual data points are denoted as x_i, this set includes the individual road segments along with the corresponding labels. In this context, the individual data points are defined as a combination of a tri-axial accelerometer signal $A_i \in \mathbb{R}^{(T\times 3)}$ sampled at a frequency $f_s \in \{50, 100\}$ Hz and an RGB image frame $I_i \in \mathbb{R}^{(H\times W\times 3)}$. In addition, the data points are provided with a corresponding label denoted as $y_i \in \{0, 1, 2\}$, where the individual labels are defined as Normal, Speed Breaker, and Pothole. This problem has been considered as a flat three-class classification problem as per the existing literature, which is defined as:

$$f_flat : (A_i, I_i) \rightarrow \hat{y}_i \in \{0, 1, 2\}$$

However, this problem has been considered as a cascaded classification problem consisting of two binary classification problems in the proposed framework.

B. Cascaded Binary Classification Architecture

Stage 1 - Hazard Presence Detection: {y_S1}

y_S1 = { 0, if y_i = 0 (Normal); 1, if y_i > 0 (Hazard) }

Stage 2 - Hazard Type Discrimination {y_S2} - to be used only if {y_S1} = 1.

y_S2 = { 0, if y_i = 1 (Speed Breaker); 1, if y_i = 2 (Pothole) }

Prediction Function: $\hat{y}_i$ = { 0 (Normal) if y_S1 = 0; 1 (Speed Breaker) if y_S1 = 1, y_S2 = 0; 2 (Pothole) if y_S1 = 1,

C. Multi-Modal Probabilistic Late Fusion

We are calculating two individual probabilities, P_sensor, which is obtained from the time-series classifier of TensorFlow, and P_vision, which is obtained from the confidence score of YOLOv8.

Probability of taking a final decision: P_final = α * P_sensor + (1 - α) * P_vision, where 0 <= α <= 1.

The optimal fusion weight, α = 0.6, was found empirically through pilot experiments on a held-out validation set, where more confidence was assigned to the accelerometer modality based on its temporal consistency and robustness to occlusion, compared to vision-based inference under changing illumination conditions. Sensitivity analysis of α is reserved for future work.

D. *Accelerometer preprocessing pipeline*

The raw tri-axial accelerometer reading \{a_x, a_y, a_z\} is preprocessed through the following steps: (1) signal magnitude area (sma) smoothing using the sliding window approach with a window of size w=10; (2) spike detection based on the condition that the magnitude of the smoothed vertical acceleration is greater than \theta = k * \mu, where \mu is the moving average of the last 2s and k=2.5; (3) fixed-length segments of length t samples containing the spike event, standardized to have zero mean and unit variance prior to the classifier submission.

E. *YOLOv8 visual inference pipeline*

The visual frames trigger based on the spike event and are then fed into the YOLOv8 model with the c2f backbone, pan neck, and decoupled detection head with the anchor-free option, which is fine-tuned on the dataset. The maximum class confidence is extracted and normalized over the target-class detections to obtain p_vision. In cases where the frames are irrelevant, p_vision is set to 0.0 to allow the sensor modality to make independent decisions.

IV. SYSTEM ARCHITECTURE AND MODEL DESIGN

A. *Full-Stack System Overview*

The RoadGuard-AI system architecture is divided into three vertically integrated layers: mobile data acquisition, cloud inference and persistence, and geospatial visualization. Data is collected from sensors via secure RESTful API calls, processed in the cloud, and visualized in near real-time on an interactive web interface.

B. *Mobile Data Acquisition Layer*

The mobile client is built using React Native 0.81, utilizing the Expo framework and Expo Sensors API for accelerometer data capture at 100 Hz or greater, along with a state machine controlling initialization, segmenting, and transmission via RESTful APIs. Timestamped GPS coordinates are sent along with each extracted segment for geospatially aware hazard assignment.

C. *Cloud Inference and Persistence Layer*

The backend inference microservice uses FastAPI, the runtime stack used was Python 3.11 3.10+, TensorFlow 2.20, PyTorch 2.10, and Ultralytics YOLOv8. The application stack used was FastAPI, SQLAlchemy/SQLite, React 19 with Vite and TailwindCSS, and React Native 0.81 with Expo Sensors. It provides REST endpoints for ingesting events, dispatching model inferences, and querying historical data. Upon receipt of the trigger message, the microservice utilizes parallel inference in classifying the hazard events by concurrently sending the accelerometer segment to the TensorFlow 2.20 time series classifier and the image frame to YOLOv8 through the Ultralytics Python 3.11 API. The classified hazard event is then persisted to the SQLite database by using the SQLAlchemy API. Additionally, spatial-temporal deduplication logic is also introduced to filter out duplicate events within a specified spatial radius and temporal window.

D. *Geospatial Visualization Layer*

The frontend dashboard is based on React 19 as the underlying technology stack, and it is augmented with Vite and TailwindCSS technologies to enable rapid development and UX/UI design. The detected hazard events in the scene are visualized using the leaflet.js library, which is used for the geospatial visualization of the hazard events. The hazard events are visualized as color-coded markers on the map, where blue color is for

Normal events, Amber color is for Speed Breaker events, and Red color is for Pothole events. Moreover, the hazard metadata is also provided in tabular format, including the timestamp, GPS coordinates, confidence score, and probabilities for all the modalities.

E. Algorithm

Algorithm: Cascaded Inference Pipeline

Algorithm : RoadGuard-AI Cascaded Inference

Input: A (accel. segment), I (image), α, K

Output: $\hat{y} \in \{0=\text{Normal}, 1=\text{SpeedBreaker}, 2=\text{Pothole}\}$

1: SMA_A ← smooth(A, w=10)

2: IF max(SMA_A) ≤ K*mean(SMA_A) THEN return 0 {No spike detected → Normal} {Spike confirmed: max(SMA_A) > K*mean(SMA_A) → continue}

A spike is confirmed when the magnitude exceeds $\theta = k \times \mu$. If no spike is detected, the segment is immediately classified as Normal and processing halts.

3: P_sensor ← TF_Classifier(normalize(SMA_A))

4: P_vision ← YOLOv8_Confidence(I)

5: P_final ← α*P_sensor + (1-α)*P_vision

6: y_S1 ← Stage1_Classifier(P_final)

7: IF y_S1 = 0 THEN return 0

8: y_S2 ← Stage2_Classifier(P_final)

9: return 1 if y_S2=0 else return 2

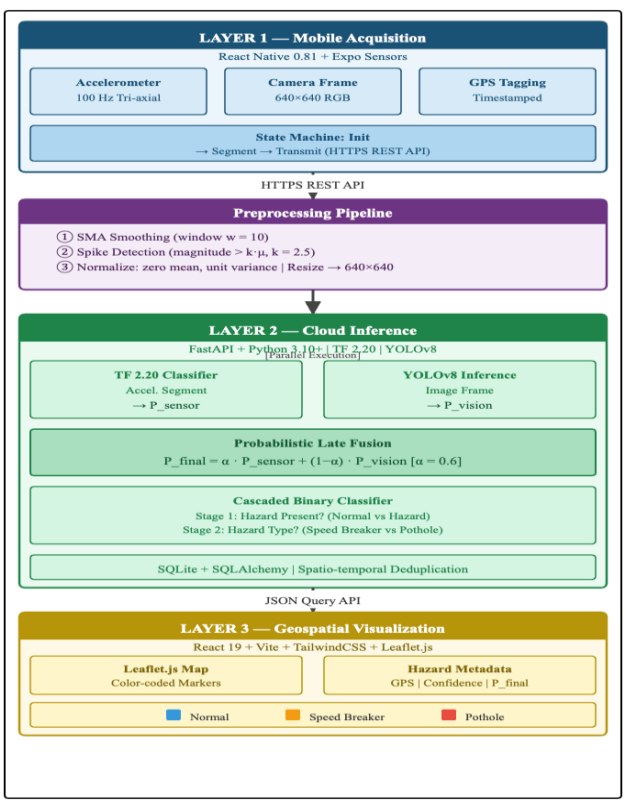

Fig. 1. RoadGuard-AI full-stack system architecture comprising three vertically integrated layers: mobile acquisition (React Native 0.81), cloud inference (FastAPI, TensorFlow 2.20, YOLOv8), and geospatial visualization (Leaflet.js).

V. EXPERIMENTAL SETUP

A. Dataset Description

The evaluation dataset comprises 874 samples collected in-house using the RoadGuard-AI mobile client. Class distribution: Normal (Class 0): 161 samples (18.4%), Speed Breaker (Class 1): 678 samples (77.6%), Pothole (Class 2): 35 samples (4.0%). Tri-axial accelerometer signals were collected at fs=50Hz. Segment length T=100 samples (2.0 seconds). Stratified 5-Fold cross-validation (K=5) was employed for evaluation.

B. *Preprocessing and Feature Engineering*

Accelerometer time series segments of a fixed length T were extracted from spikes triggered activation. The segments were standardized by centering at zero mean and unit variance for each accelerometer channel. Image frames were resized to a fixed size of 640 x 640 pixels for YOLOv8 model compatibility. Two binary flags for data augmentation were tested in the ablation study: (1) class weighting: inverse frequency loss weight per class; (2) data augmentation: horizontal flipping, time shift, and Gaussian noise injection for training fold samples only.

C. *Evaluation Protocol*

The model's performance was measured by stratified K-Fold cross-validation for accuracy, precision, recall, F1-score calculation by using scikit-learn library. A paired two-tailed t-test was used for statistical comparison of cascaded and flat model architectures by comparing per-fold accuracy values for significance threshold $p < 0.05$.

D. *Implementation details*

All the models were trained using the adam optimizer and the following loss functions: binary cross-entropy was used as the loss function for the stage 1 classifier and the stage 2 classifier, and categorical cross-entropy was used as the loss function for the flat three-class model.the runtime stack used was python 3.10+, tensorflow 2.20, pytorch 2.10, and ultralytics YOLOv8.the application stack used was FastAPI, sqlalchemy/sqlite, react 19 with vite and tailwindcss, and react native 0.81 with expo sensors.

VI. RESULTS AND ANALYSIS

A. *Cross-validation performance comparison*

Table i summarizes the cross-validation accuracy results for the flat three-class model and the proposed cascaded architecture model. The proposed cascaded architecture has a cross-validation accuracy of 0.551, which is a relative improvement of 20.3% over the flat three-class model's cross-validation accuracy of 0.458.

TABLE I

Cross-Validation Mean Accuracy: Cascaded vs. Flat

Classification Architecture	CV Mean Accuracy
Flat Three-Class Baseline	0.458
Cascaded Binary (Proposed)	0.551

A paired t-test on the values for per-fold accuracy revealed a p-value of 0.107, which does not attain statistical significance at a threshold α=0.05. This implies both architectures are statistically equivalent in terms of cross-validation accuracy on these tested folds, making them both worthy of further exploration.

The lack of statistical significance is attributed to the limited dataset size (n=874) and severe class imbalance (4% Pothole class). Future work will address this through expanded data collection targeting underrepresented classes.

B. *Ablation Study*

Table II shows the ablation study results of the models over four different configurations.

TABLE II

Ablation Study: Regularization Effects on Architecture Performance

Config.	Wts.	Aug.	Casc. Acc.	Flat Acc.
No Wts, No Aug	No	No	0.725	0.673
No Wts, Aug	No	Yes	0.689	0.764
Wts, No Aug	Yes	No	0.488	0.491
Wts + Aug	Yes	Yes	0.499	0.504

As shown, the cascaded architecture attains the highest accuracy of 0.725 under the no weights, no augmentation condition, validating the effectiveness of stage-specific optimization of boundaries over the natural class distribution. On the other hand, data augmentation has a more significant effect on the flat architecture, with a difference of 0.764 compared to 0.689 under the no augmentation condition implying that the induced distribution shift affects the smaller cascaded training sets more adversely. The effect of class weighting, on the other hand, appears to be adverse, as indicated by a reduction in accuracy over both architectures, likely due to over-penalization of majority class samples during small training.

C. *Label encoding defect identification*

In the course of conducting initial experiments, an anomalously low cv mean accuracy of 0.188 was noted, which was below the random baseline for three classes. Systematic debugging of the problem revealed a label encoding defect in the cross-validation loop, where class integer labels were remapped anomalously between stage 1 and stage 2 relabeling operations. After correction, the cascaded architecture was used to generate the results presented in tables i and ii.

D. *Statistical significance analysis*

The t-test was performed, and the result was p = 0.107. Although the cascaded architecture shows a directional advantage, this is still within the sampling variance of the fold level, again pointing to the need for additional evaluation data sets to prove the superiority of the architecture.

VII. DISCUSSION

The implemented system was validated through a full-stack prototype comprising a React Native mobile data acquisition layer operating at 50Hz accelerometer sampling, a FastAPI cloud inference backend with parallel TensorFlow and YOLOv8 inference, spatial-temporal deduplication logic (50m radius, 60s window), and a Leaflet.js geospatial visualization dashboard. The complete system demonstrates end-to-end deployability for pilot-scale civic road monitoring programs.

The results obtained from the experiment validate the complex nature of the cascaded binary decomposition strategy. The best performance is obtained when the cascaded structure is used without any form of class weighting and data augmentation, as the binary classifiers can optimize the decision boundaries on the natural class distribution. This is in accordance with the motivation behind cascaded classifiers, as the binary discrimination problem is inherently easier and has tighter decision boundaries compared to the three-class problem with its complex and overlapping decision manifold.

The decrease in accuracy with the addition of class weighting is unexpected and could be due to the over-penalization of the majority class samples in the training set folds. A direction for future work is the incorporation and evaluation of oversampling methods such as SMOTE as an alternative to loss-level weighting.

The fixed weight for fusion, α = 0.6, was empirically determined depending on sensor reliability in normal circumstances. Adaptive calibration of α depending on vehicle speed, ambient luminance, and sensor quality indicators is another promising design option. The restrictions associated with continuous background sensing on iOS platforms are another issue to be taken into account,

which may be overcome by hardware offloading using sensing pods based on Bluetooth Low Energy technology. The current SQLite-based persistence layer, though adequate for the prototype-scale evaluation, poses a scalability bottleneck under scenarios of concurrent high-frequency multi-device ingestion operations. Architectural migration strategies, along with their performance considerations, are left as future work.

VIII. CONCLUSION AND FUTURE WORK

The paper introduced a novel hybrid edge-cloud multi-modal framework for road hazard detection in real-time, fusing accelerometer time-series analysis with YOLOv8 computer vision inference using weighted probabilistic late fusion. The key architectural innovation of the paper is the introduction of a hierarchical two-stage cascaded binary classification framework to detect road conditions in three classes by splitting the problem into hazard presence detection and hazard type detection sub-problems. The cascaded framework achieves directional improvement of 20.3% relative ($p=0.107$), warranting further evaluation with a larger dataset in mean cross-validation accuracy to 0.551, whereas the maximum accuracy of 0.725 is achieved by the baseline configuration.

A detailed ablation study on these interactions showed a nuanced relationship between class weighting, data augmentation, and classification architecture type. Correcting a label encoding flaw is a further example of the need for careful experimental verification in hierarchical classification systems. The complete stack deployment architecture represents a deployable solution for a civic ai platform for pilot-level road monitoring programs.

Future work: we plan to extend this work in the following directions: (1) tensorflow lite quantization for edge inference on mobile devices; (2) postgresql with postgis migration for municipal-level spatial queries; (3) adaptive spike detection threshold instead of a fixed k coefficient; (4) severity regression head for continuous hazard prediction; (5) large-scale real-world deployment for cross-domain generalization over varied road types and geographical regions.

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